

JEE Main - 16 | JEE - 2022

Date: 25/12/2021

Maximum Marks: 300

Timing: 09:00 AM - 2:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

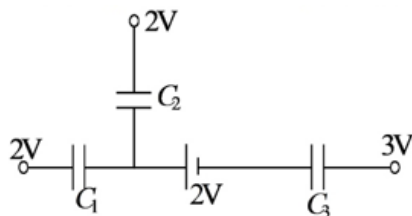
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

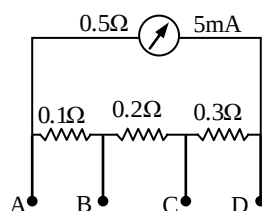
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A part of the circuit is shown in the figure. All the capacitors have capacitance of $2\mu F$. Then :



- (A) Charge on capacitor C_1 is zero
 (B) Charge on capacitor C_2 is zero
 (C) Charge on capacitor C_3 is zero
 (D) None of these
2. A milli ammeter of range 5 mA and resistance 0.5Ω is joined in a circuit as shown to form an ammeter. Find the value of current entering A for which meter gives full scale deflection when A and B are used as terminals :

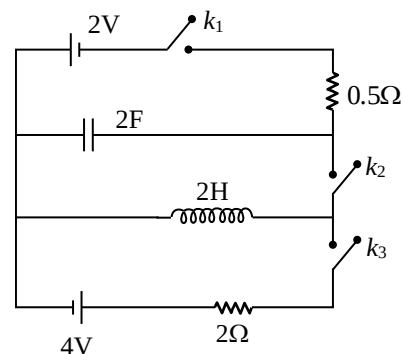


- (A) 50 mA
 (B) 5 mA
 (C) 55 mA
 (D) 9.2 mA
3. Two equal point charges are fixed at $x = -a$ and $x = +a$ on the x -axis. Another point charge Q is placed at the origin. The change in the electrical potential energy of system, when it is displaced by a small distance x along the x -axis, is approximately proportional to :
- (A) x
 (B) x^2
 (C) x^3
 (D) $\frac{1}{x}$
4. A charge particle enters into a region containing uniform electric field (E) and uniform magnetic field (B) along x -axis and y -axis, respectively. If it passes the region un-deviated, the velocity of charge particle may be given by :

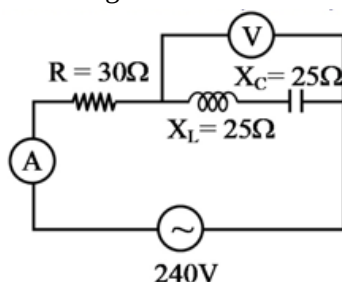
- (A) $2\hat{i} + \frac{E}{B}\hat{k}$
 (B) $2\hat{j} + \frac{E}{B}\hat{k}$
 (C) $-\frac{E}{B}\hat{k}$
 (D) None of these

5. For the circuit shown here, keys k_1 and k_3 are closed for 1 second. Now keys k_1 and k_3 are opened and key k_2 is closed at the same instant. Maximum charge on the capacitor after key k_2 is closed is :

- (A) $4\left(1 - \frac{1}{e}\right)C$
 (B) $4\sqrt{2}\left(1 - \frac{1}{e}\right)C$
 (C) $8\left(1 - \frac{1}{e}\right)C$
 (D) Zero

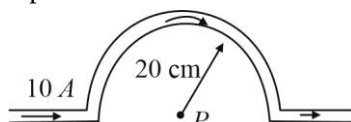


6. In the following circuit the readings of AC voltmeter and ammeter will be respectively :



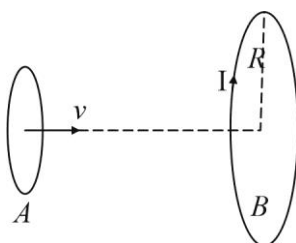
- (A) 0V, 3A (B) 150V, 3A (C) 150V, 6A (D) 0V, 8A

7. A current of $10A$ is passing through a long wire which has semicircular loop of the radius 20 cm as shown in the figure. Magnetic field produced at the centre of the loop is :



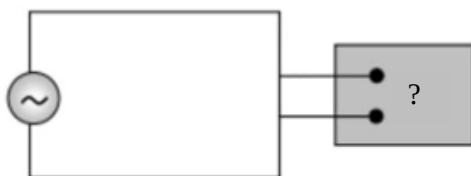
- (A) $10\pi\mu T$ (B) $5\pi\mu T$ (C) $4\pi\mu T$ (D) $2\pi\mu T$

8. Loop A of radius r ($r \ll R$) moves towards a constant current carrying loop B with a constant velocity v in such a way that their planes are parallel and coaxial. The distance between the loops when the induced EMF in loop A is maximum is :

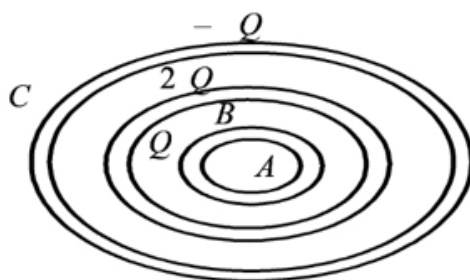


- (A) R (B) $\frac{R}{\sqrt{2}}$ (C) $\frac{R}{2}$ (D) $R\left(1 - \frac{1}{\sqrt{2}}\right)$

9. Following figure shows an ac generator connected to a “black box” through a pair of terminals. The box contains possible R , L , C or their combination, whose elements and arrangements are not known to us. Measurements outside the box reveals that $\varepsilon = 75 \sin(\omega t)$ volt, $i = 1.5 \sin(\omega t + 45^\circ)$ amp, then the wrong statement is :



- (A) There must be a capacitor in the box (B) There must be an inductor in the box
(C) There must be a resistance in the box (D) The power factor is 0.707
10. Charges Q , $2Q$ and $-Q$ are given to three concentric conducting spherical shells A, B and C respectively, as shown in the figure. The ratio of charges on the inner and the outer surfaces of the shell C will be :

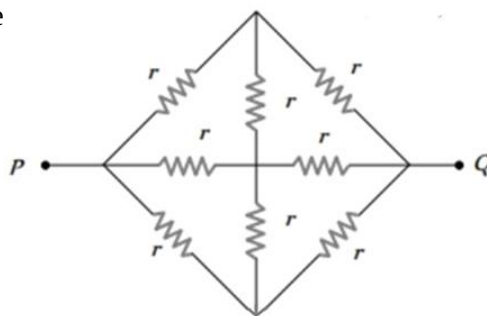


- (A) $+\frac{3}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$
11. The electric field $\vec{E}$ is given by $\vec{E} = a\hat{i} + b\hat{j}$ (where a and b is constant and $\hat{i}$, $\hat{j}$ are unit vector along x and y axis respectively), the flux passing through a square area of side l and parallel to y - z plane is :
- (A) bl^2 (B) al^2 (C) $\sqrt{(a^2 + b^2)}l^2$ (D) $\sqrt{(a^2 - b^2)}l^2$
12. A spherical drop of capacitance $1 \mu F$ is broken into 8 drops of equal radius. Then, the capacitance of each small drop is :
- (A) $\frac{1}{2} \mu F$ (B) $\frac{1}{4} \mu F$ (C) $\frac{1}{8} \mu F$ (D) $8 \mu F$
13. A uniform solid non conducting sphere of radius R and total charge q rotates about its diametric axis with constant angular speed ω . The magnetic moment of the sphere is :
- (A) $\frac{1}{3}qR^2\omega$ (B) $\frac{2}{3}qR^2\omega$ (C) $\frac{1}{5}qR^2\omega$ (D) $\frac{2}{5}qR^2\omega$
14. A battery of emf E and internal resistance r is connected to a resistor of resistance r_1 and Q joules of heat is produced in r_1 in a certain time t . When the same battery is connected to another resistor of resistance r_2 , the same quantity of heat is produced in r_2 in the same time t , the value of r is :

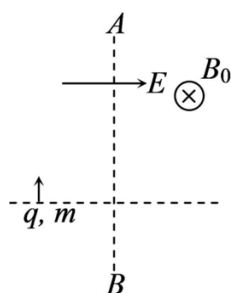
- (A) $\frac{r_1^2}{r_2}$ (B) $\frac{r_2^2}{r_1}$ (C) $\frac{1}{2}(r_1 + r_2)$ (D) $\sqrt{r_1 r_2}$

15. The equivalent resistance between the points P and Q in the network given here is equal to : (Given $r = \frac{3}{2}\Omega$)

- (A) $\frac{1}{2}\Omega$ (B) 1Ω
(C) $\frac{3}{2}\Omega$ (D) 2Ω

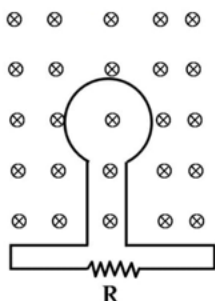


16. A uniform electric field E is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line AB as shown. A charge particle having charge q and mass m is projected vertically upward and crosses the line AB after time t_0 . Find the speed of projection if particle moves after t_0 with constant velocity. (Given $qE = mg$)



- (A) gt_0 (B) $2gt_0$ (C) $\frac{gt_0}{2}$
(D) particle can't move with constant velocity after crossing AB

17. In the figure the flux through the loop perpendicular to the plane of the coil and directed into the paper is varying according to the relation $\phi = 6t^2 + 7t + 1$ where ϕ is in milliweber and t is in second. The magnitude of the emf induced in the loop at $t = 2s$ and the direction of induce current through R are :



- (A) 39 mV, right to left (B) 39 mV, left to right
(C) 31 mV, right to left (D) 31 mV, left to right

18. If an electron enters into a space between the plates of a parallel plate capacitor at an angle α with the plates and leaves at an angle β to the plates. The ratio of it's kinetic energy while entering the capacitor to that leaving will be :

- (A) $\left(\frac{\cos \alpha}{\cos \beta}\right)^2$ (B) $\left(\frac{\cos \beta}{\cos \alpha}\right)^2$ (C) $\left(\frac{\sin \alpha}{\sin \beta}\right)^2$ (D) $\left(\frac{\sin \beta}{\sin \alpha}\right)^2$

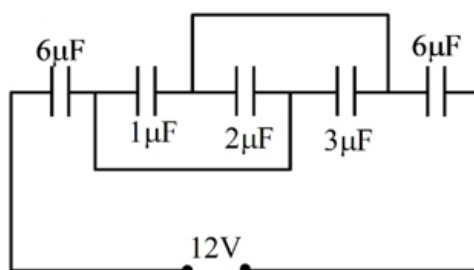
19. An inductor L and a resistor R are connected in series with a direct current source of emf E . The maximum rate at which energy is stored in the magnetic field is :
- (A) $\frac{E^2}{4R}$ (B) $\frac{E^2}{R}$ (C) $\frac{4E^2}{R}$ (D) $\frac{2E^2}{R}$
20. A tightly-wound long solenoid has n turns per unit length, radius r and carries a current i . A particle having charge q and mass m is projected from a point on the axis in the direction perpendicular to the axis. The maximum speed for which particle does not strike the solenoid will be :
- (A) $\frac{\mu_0 q r n i}{2m}$ (B) $\frac{\mu_0 q r n i}{m}$ (C) $\frac{2\mu_0 q r n i}{3m}$ (D) None of these

SECTION-2

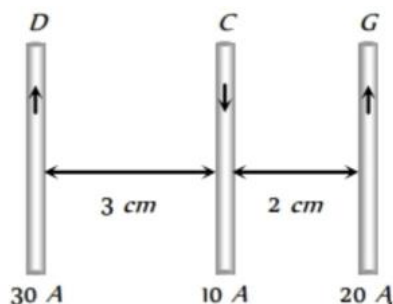
This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

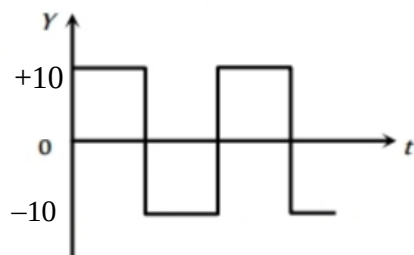
21. Figure shows in five capacitors connected across a 12 V power supply. What is the potential drop (in volts) across the $2\mu F$ capacitor ?



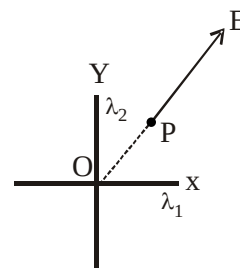
22. A factory is served by 220 volt supply line. In a circuit protected by a fuse marked 10 ampere, the maximum number of 100 watt lamps in parallel that can be turned on is _____.
23. Three long, straight parallel wires carrying current, are arranged as shown in figure. The force experienced by a 25 cm length of wire C is _____ (in Newtons).



24. The r.m.s. voltage of the wave form shown is _____ (in volts).



25. Two mutually perpendicular infinite wires along x -axis and y -axis carry linear charge densities λ_1 and λ_2 respectively. The electric line of force at P is along the line $y = \frac{1}{\sqrt{3}}x$, where P is also a point lying on the same line then find λ_2 / λ_1



PART II : CHEMISTRY**100 MARKS****SECTION-1**

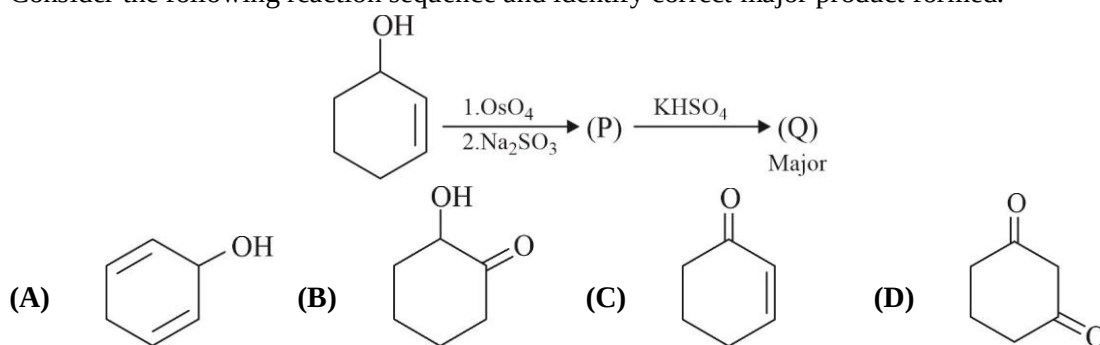
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which of the following is TRUE regarding structure of crystalline $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?

- I. It contains coordinated and interstitial water
- II. It contains coordinated and H-bonded water
- III. Its unit cell is symmetrical ($a = b = c$; $\alpha = \beta = \gamma = 90^\circ$)
- IV. Its unit cell is not symmetrical ($a \neq b \neq c$; $\alpha \neq \beta \neq \gamma \neq 90^\circ$)

(A) only II (B) only III (C) II, III (D) II, IV

2. Consider the following reaction sequence and identify correct major product formed.



3. Which of the following is NOT TRUE for amino acid, methionine?

- (A) In Lassaigne's test it produce blood red colour
- (B) It can't be synthesized in the body
- (C) It has one asymmetric carbon atom and its configuration is D
- (D) In zwitter ionic form, it show amphoteric behaviour

4. The concentration of an aqueous solution of non volatile monoprotic acid measured first by freezing point depression and then by boiling point elevation. The solution is found to be 0.83m by freezing point depression and to be 0.72m by boiling point elevation. Which is the best explanation for this discrepancy?

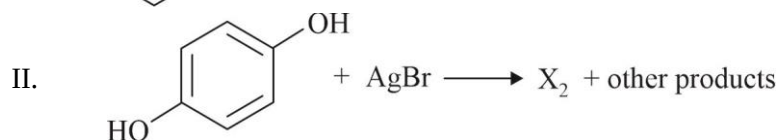
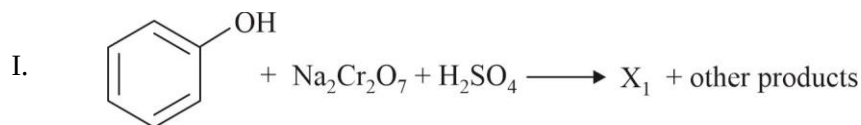
- (A) Acid associates partially into dimers at lower temperatures
- (B) The boiling point elevation constant for water is smaller than its freezing point depression constant
- (C) The volume of solution is greater at higher temperature
- (D) Ionization of the acid is markedly exothermic

5. Which of the following reaction(s) is correctly represented for the preparation of chlorine?

- (A) $\text{NaCl} + \text{H}_2\text{SO}_4 + \text{MnO}_2 \longrightarrow$ (B) $\text{NH}_3 + \text{CaOCl}_2$
- (C) $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow$ (D) $\text{NaCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$

6. Non-stoichiometric cuprous sulphide, Cu_2S can be prepared in laboratory. In this sulphide copper to sulphur ratio is slightly less than 2 : 1. This substance is:
- (A) diamagnetic (B) a p-type semiconductor
(C) a n-type semiconductor (D) a perfectly crystalline solid

7. Consider reaction I and II.



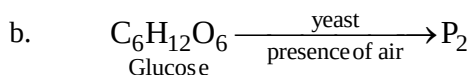
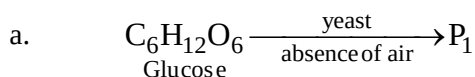
X_1 and X_2 are organic compounds. Identify correct statement.

- (A) X_1 and X_2 are identical
(B) X_1 and X_2 are different
(C) X_1 and X_2 are constitutional isomers
(D) X_1 and X_2 are canonical forms
8. The correct combination of the main forces which stabilize the 2° and 3° structures of proteins is:
- (A) Covalent bond, electrovalent bond, hydrogen bond, peptide linkages
(B) Hydrogen bond, electrovalent bond, van der Waal's forces, peptide linkages
(C) Hydrogen bond, disulphide linkages, Van der Waal's forces, electrostatic forces of attraction
(D) Peptide linkages, disulphide linkages, Van der Waal's forces, electrostatic forces of attraction.
9. A solution having w g oxalic acid ($M_o = 90 \text{ g/mole}$) in 100g H_2O has a freezing point of -1.3°C . If 3g sodium hydroxide ($M_o = 40 \text{ g/mole}$) is added to above solution then pH of the solution will be. Assume molality $\approx$ molarity, $K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C/m}$, for oxalic acid $\text{pK}_{a1} = 1.27$ & $\text{pK}_{a2} = 4.28$, $\alpha_1 = 0.4$, $\log 3 = 0.47$
- (A) 1.27 (B) 4.28 (C) 2.775 (D) 9.10
10. Which of the following reaction is not disproportionation reaction?
- (A) $\text{Br}_2(\ell) + \text{Na}_2\text{CO}_3(\text{aq}) \longrightarrow \text{Products}$
(B) $\text{XeF}_6(\text{s}) + \text{NaOH}(\text{aq}) \longrightarrow \text{Products}$
(C) $(\text{CN})_2(\text{g}) + \text{KOH}(\text{aq}) \longrightarrow \text{Products}$
(D) $\text{Br}_2(\ell) + \text{Na}_2\text{S}_2\text{O}_3(\text{aq}) \longrightarrow \text{Products}$
11. A solid calcium boride CaB_x has a density of 2.5 g/cc . It has a primitive lattice with unit cell parameters, $a = b = c = 415 \text{ pm}$, $\alpha = \beta = \gamma = 90^\circ$ and one calcium atom per unit cell. What is the value of X?

(Atomic mass $\text{Ca} = 40$, $\text{B} = 10.91$, $N_A = 6 \times 10^{23}$)

- (A) 8 (B) 4 (C) 6 (D) 2

12. Consider reaction (a) and (b).



Which of the following is correct?

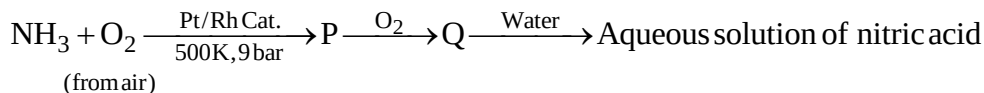
- (A) Both P_1 and P_2 shows Tollen's reagent test
 (B) Both P_1 and P_2 shows iodoform test
 (C) Both P_1 and P_2 shows hydrogen bonding
 (D) Both P_1 and P_2 shows 2, 4-DNP test

13. Which of the following is correct match?

I	Vitamin B ₁₂	(P)	Increased fragility of RBCs
II	Vitamin E	(Q)	Increased blood clotting time
III	Vitamin K	(R)	RBC deficient in haemoglobin
		(S)	Bleeding gums

- (A) I – P; II – Q, R ; III - S
 (B) I – Q, S; II – P ; III – R
 (C) I – R; II – P ; III - Q
 (D) I – P; II –R ; III - S

14. On large scale, nitric acid is prepared by Ostwald's process.



Which of the following is NOT correct?

- (A) Intermediate products P and Q are paramagnetic
 (B) Dissolution of 1L pure liquid nitric acid with 1L pure water results in formation of 2L liquid solution
 (C) Solution of nitric acid and water can't be separated completely into pure nitric acid and pure water by distillation
 (D) Equimolar mixture of P and Q on cooling to 250 K form blue coloured solid

15. Which of the following is correct match?

I	F	a	Exhibits only one nonzero oxidation state
II	Ne	b	Exhibits only positive oxidation state
III	I	c	Exhibits both positive and negative oxidation state
IV	Xe	d	Exhibits neither the negative oxidation state nor does the positive oxidation state

- (A) I – c ; II – d ; III – c ; IV - b
 (B) I – a ; II – b ; III – c ; IV – c
 (C) I – a ; II – d ; III – b ; IV - c
 (D) I – a ; II – d ; III – c ; IV - b

16. Which of the following is correct match?

I.	LiH	a.	Ionic solid
II.	CH ₄	b.	Covalent - solid
III.	H ₂ O	c.	Molecular solid
IV.	Xe	d.	H bonded molecular solid

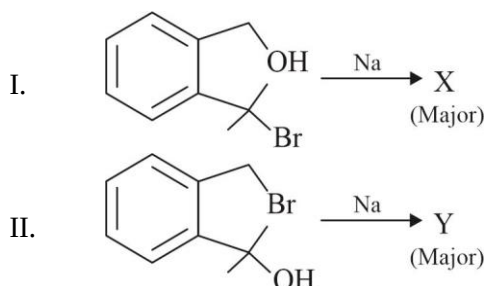
(A) I – d; II – b; III – a ; IV - c

(B) I – a; II – b; III – d ; IV – c

(C) I – a; II – c; III – d ; IV - c

(D) I – a; II – c; III – c ; IV - b

17. Consider reaction I and II.



Identify correct statement regarding X and Y.

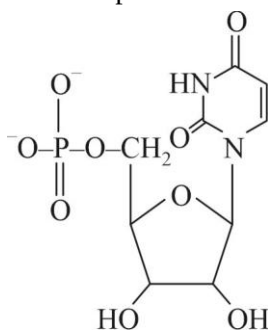
(A) X and Y are diols

(B) X and Y have same boiling point

(C) X in presence of H⁺ catalyst gives Y

(D) X and Y are not isomers

18. The structure shown in the diagram is an example of a monomer unit used in the formation of:



(A) RNA

(B) Protein

(C) DNA

(D) Polysaccharide

19. 800g pure acetone ($d = 0.8 \text{ g/cc}$, $M_o = 58 \text{ g/mole}$, $T = 25^\circ\text{C}$) is mixed with 1500 g pure chloroform ($d = 1.5 \text{ g/cc}$, $M_o = 120 \text{ g/mole}$, $T = 25^\circ\text{C}$) to prepare a liquid solution.

(Given: Atomic mass, H = 1, C = 12, O = 16, Cl = 35.67, K = 39)

(A) Mixing results in formation of two layers, lower layer is of chloroform and upper layer is of acetone

(B) The volume of resulting liquid solution is two litre

(C) Mixing of acetone and chloroform is endothermic process

(D) Addition of 56 gm KOH results in formation of 2.225 kg white solid.

20. Which of the following is correct order of S-S bond length?

(A) Monoclinic sulphur form > cyclo-S₆ chair form > Diatomic sulphur vapour form

(B) Diatomic sulphur vapour form > Monoclinic sulphur form > cyclo- S₆ chair form

(C) Cyclo- S₆ chair form > Monoclinic sulphur form > Diatomic sulphur vapour form

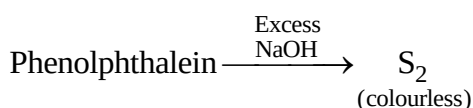
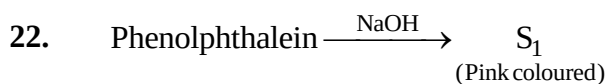
(D) Diatomic sulphur vapour form > cyclo-S₆ chair form > Monoclinic sulphur form

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

21. The structure of unit cell of perovskite a salt of lanthanum (La), manganese (Mn) and oxygen (O), has Mn^{2+} at each corner, oxide on every edge centre and lanthanum ion at the body centre of cubic unit cell. The sum of coordination number of Mn^{2+} , Li^{n+} and O^{2-} is _____.



Find the sum of degree of unsaturation of compound S_1 and compound S_2 .

23. Insulin is a polypeptide having 'X' amino acids and have a well defined conformation of a protein. The value of X is _____.
24. A current of dry air was passed first through a series of bulbs containing a solution of $\text{C}_6\text{H}_5\text{NO}_2$ in ethanol of molality 0.725 m and then through a series of bulbs containing pure ethanol ($T = 284\text{K}$). The loss in weight of the solvent bulb was 0.0685 g. Calculate the loss in weight (in gm) of the solution bulbs. (Given: Atomic mass, H = 1, C = 12, N = 14, O = 16,)
25. How many of the following statements are correct regarding structure and properties of S_2F_2 , S_2F_4 , S_2F_{10} , SF_4 and SF_6 ?
- S_2F_{10} exists in structure in which each sulphur atom of the S_2F_{10} molecule is octahedral and surrounded by five fluorine atoms
 - S_2F_2 undergoes disproportionation as shown by reaction. $2\text{S}_2\text{F}_2 \longrightarrow \text{SF}_4 + 3\text{S}$
 - S_2F_{10} on reaction, with XeF_2 undergoes redox reaction as shown by reaction,
 $\text{S}_2\text{F}_{10} + \text{XeF}_2 \longrightarrow 2\text{SF}_6 + \text{Xe}$
 - SF_4 and SF_6 , both are gaseous in nature and SF_4 is less volatile than SF_6
 - SF_6 is exceptionally stable for steric reasons

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If for $x > 0$, $f(x) = (a - x^n)^{1/n}$, $g(x) = x^2 + px + q$, $p, q \in R$ and the equation $g(x) - x = 0$ has imaginary roots, then the number of real roots of equation $g(g(x)) - f(f(x)) = 0$ is :
 (A) 0 (B) 2 (C) 4 (D) n
- $f(x) = \{x\} + \{x + 1/3\} + \{x - 1/3\}$ where $\{x\}$ denotes the fractional part of x . Find fundamental period of $f(x)$:
 (A) $1/3$ (B) $2/3$ (C) 1 (D) None of these
- $f : R \rightarrow \left(0, \frac{\pi}{2}\right]$, $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$. Find complete values of α for which $f(x)$ is onto :
 (A) $\left[\frac{1 - \sqrt{17}}{2}, \frac{1 + \sqrt{17}}{2}\right]$ (B) $\frac{1 \pm \sqrt{17}}{2}$
 (C) $\left(-\infty, \frac{1 - \sqrt{17}}{2}\right] \cup \left[\frac{1 + \sqrt{17}}{2}, \infty\right)$ (D) None of these
- The range of the function $f(x) = \sin^{-1}(\log_e[x]) + \log_e(\sin^{-1}[x])$ where $[]$ denotes the greatest integer function is :
 (A) R (B) $[1, 2)$ (C) $\{\log_e(\pi/2)\}$ (D) $\{-\sin 1\}$
- $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2} \cos x \cos 3x}{x^2}$ is equal to :
 (A) $11/2$ (B) $7/2$ (C) $9/2$ (D) None of these
- If $\lim_{x \rightarrow 0} \frac{\log_e \cot\left(\frac{\pi}{4} - k_1 x\right)}{\tan k_2 x} = 1$
 (A) $k_1 = k_2$ (B) $2k_1 = k_2$ (C) $k_1 = 2k_2$ (D) $k_1 = 4k_2$
- $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = a$, $\lim_{x \rightarrow 0} \frac{f(1 - \cos x)}{g(x) \sin^2 x} = b$ ($b \neq 0$) then $\lim_{x \rightarrow 0} \frac{g(1 - \cos 2x)}{x^4}$ is :
 (A) $\frac{4a}{b}$ (B) $\frac{a}{4b}$ (C) $\frac{a}{b}$ (D) None of these

8. $f(x) = \begin{cases} 3 & , x < 0 \\ 2x+1 & , x \geq 0 \end{cases}$. Then :
- (A) Both $f(x)$ and $f(|x|)$ are differentiable at $x = 0$
 (B) $f(x)$ is differentiable but $f(|x|)$ is not at $x = 0$
 (C) $f(|x|)$ is differentiable but $f(x)$ is not differentiable at $x = 0$
 (D) Both are not differentiable at $x = 0$
9. $f(x) = (2x - \pi)^3 + 2x - \cos x$ then $\frac{d}{dx} f^{-1}(x)$ at $x = \pi$ is :
- (A) $3\pi^2 + 2$ (B) -2 (C) $1/(3\pi^2 + 2)$ (D) $1/3$
10. $f(x) = x/(1 + x \tan x)$ has in $(0, \pi/2)$
- (A) Only one minimum (B) Only one maximum
 (C) No extremum (D) More than one maximum
11. The value of m for which area of Δ included between the axes and any tangent to the curve $x^m y = b^m$ is constant is :
- (A) $1/2$ (B) 1 (C) $3/2$ (D) 2
12. Let $f(x) = \sin^{-1}(2g(x)/(1+(g(x))^2))$, then :
- (A) $f(x)$ is decreasing function if $g(x)$ is increasing and $|g(x)| > 1$
 (B) $f(x)$ is decreasing function if $g(x)$ is decreasing and $|g(x)| > 1$
 (C) $f(x)$ is an increasing function if $g(x)$ is increasing and $|g(x)| > 1$
 (D) None of these
13. Let f be a twice differential function satisfies $f(1) = 1/2$, $f(2) = 2$, $f(3) = 9/2$ then :
- (A) $f''(x) = 0$ for some $x \in (1, 3)$ (B) $f''(x) = 1$ for at least one $x \in (1, 3)$
 (C) $f''(x) = 1 \forall x \in (1, 3)$ (D) $f''(x) = 0 \forall x \in (1, 3)$
14. The minimum length of tangent drawn from point P on $3x + 4y - 20 = 0$ to circle $x^2 + y^2 = 1$ is :
- (A) 4 (B) $\sqrt{15}$ (C) $2\sqrt{2}$ (D) $\sqrt{13}$
15. Find complete set of values of k so that $\tan^{-1} x - \cos^{-1} x = k$ has a real solution :
- (A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (B) $\left[-\frac{5\pi}{2}, \frac{\pi}{2}\right]$ (C) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$ (D) $\left[-\frac{5\pi}{4}, \frac{\pi}{4}\right]$
16. If α, β, γ are roots of $x^3 - 10x + 11 = 0$. Then $\tan^{-1} \alpha + \tan^{-1} \beta + \tan^{-1} \gamma$ is :
- (A) $\frac{\pi}{4}$ (B) $\frac{5\pi}{4}$ (C) $\frac{-3\pi}{4}$ (D) None of these
17. If the tangent at $(1, 1)$ on $y^2 = x(2-x)^2$ meets the curve again at P , then P is :
- (A) $(4, 4)$ (B) $(-1, 2)$ (C) $(9/4, 3/8)$ (D) None of these

18. A, B are two square matrices such that $A^2B = BA$ and if $(AB)^{10} = A^k B^{10}$. Then k is :
 (A) 1001 (B) 1023 (C) 1024 (D) None of these
19. The equation $2 \cos^{-1} x = \sin^{-1} (2x \sqrt{1-x^2})$ is valid for all values of x satisfying :
 (A) $-1 \leq x \leq 1$ (B) $0 \leq x \leq 1$
 (C) $0 \leq x \leq 1/\sqrt{2}$ (D) $1/\sqrt{2} \leq x \leq 1$
20. $f : Q \rightarrow R, f(x) = \sin x$ is :
 (A) one-one onto (B) one-one into
 (C) many-one onto (D) many-one into

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

21. If $x_1, x_2, \dots, x_n$ are the solutions of $\cot^{-1}((x+7)(x+6)) + \cos^{-1}(x^2 + 13x + 43) = \pi/2$. Find $|x_1| + |x_2| + \dots + |x_n| + n$
22. If $x \sin(\alpha + y) = \sin y$ and $\frac{dy}{dx} = \frac{m}{x^2 + 2nx + 1}$. Then $m^2 + n^2$ is :
23. Let B be a skew symmetric matrix of order 3×3 with real entries. Given $I - B$ and $I + B$ are two non singular matrices.
 If $A = (I + B)(I - B)^{-1}$ where $\det(A) > 0$.
 Then find $\det(2A) - \det(\text{adj } A)$
24. If x, y, z are distinct even integer. Find minimum value of

$$\begin{vmatrix} 1+x^2+x^4 & 1+xy+x^2y^2 & 1+xy+x^2z^2 \\ 1+xy+x^2y^2 & 1+y^2+y^4 & 1+yz+y^2z^2 \\ 1+xz+x^2z^2 & 1+yz+y^2z^2 & 1+z^2+z^4 \end{vmatrix}$$
 is :
25. If graph of $y = f(x)$ is as shown in figure
- then number of solutions of equation $f(|x| + 1) = 1/2$ is equal to :